Session 5:

- Ssh
- Accessing Linux File Systems · mount · umount · lsblk · df · fstab
- Installing & Updating Software Packages · dnf · rpm · repos
- Archiving & Compressing Files · tar · gzip · bzip2 · xz · zip

ssh "secure shell": ch 10

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remote and secure connection to a certain server

port 22

on my lap (client)

ssh username@ip-server -p port-number

ways:

access --> password, keys(public, private) -->(client)

protocols:

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rsa, dsa, ...

1)symmetric key --> encrypt, decrypt with one key

2)assymmetric key --> encrypt, decrypt with 2 keys

default permissions --> private key == 400

--> public key == 444

client

server

public
private

signing
======
server --> challenge = 123
+
hi i am client
encrypt
challenge + message
with
client private key

== 2565&8*

server == 2565&8*
decrypt public key
of client
123
+
hi i am client

public key of client

aya

zyad

generate 2 keys
public, private

public

ahmed
mona
....

challenge:
send to challenge random 123
adn
...

private key of aya --> encrypt random challenge + message 'i am aya'
== shd;io;jdio

zyad --> shd;io;jdio + public key of aya = 123

inode:

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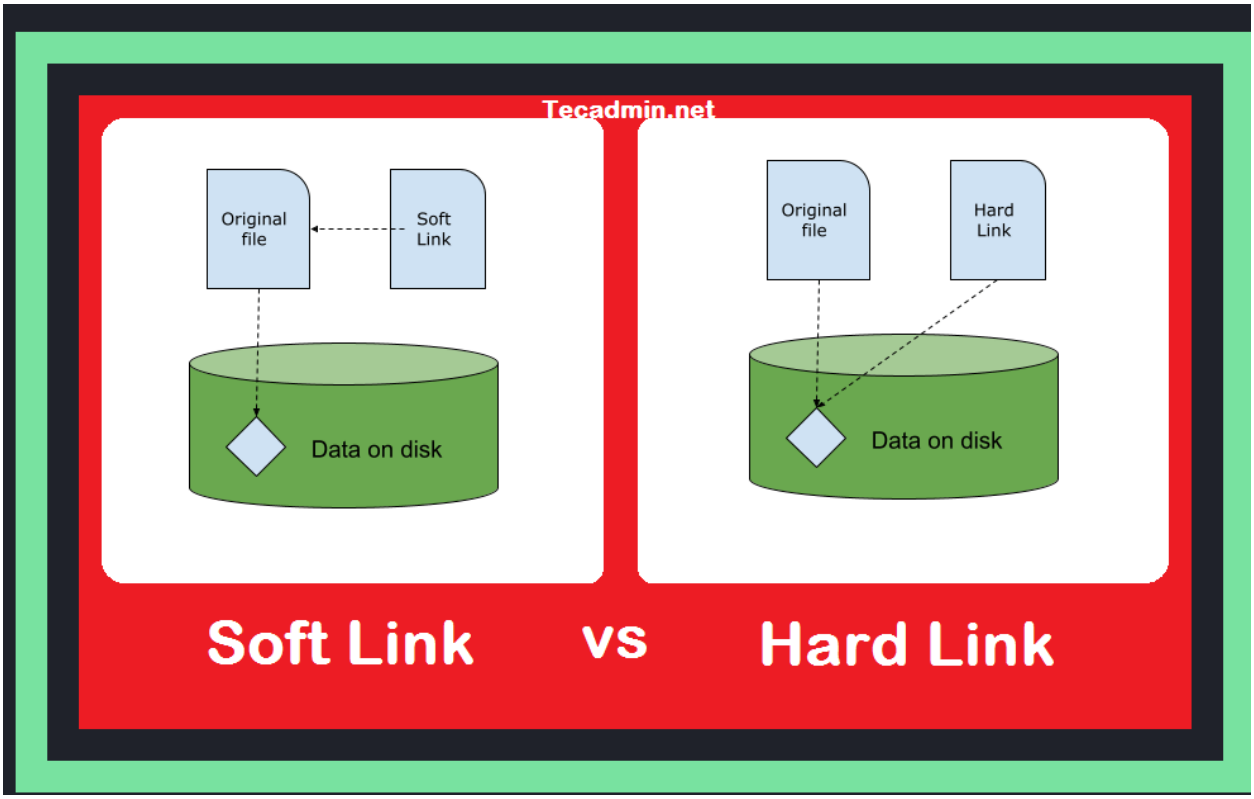
soft link (different inode)

hard link (same inode)



Storage is read --> block by block

f1 --> from block 5 to block 9

Meta data of f1 --> start from bl5

create f/dir ==> must have inode (meta data):

files:

permissions, owner, group, link, modify time,

access time, creation time, whick block ...

dir:

permissions, owner, group, link, modify time,

access time, creation time, whick block ...

+

Each file pionts to what?
Ex:
3ndy f1 --> b5

f2 --> b7

5 files 4 G

20 file 2G

40 file 50G

File vs dir.
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file --> b3--> b10
file --> Have content

directory --> special file points on other files

related to inodes
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cp vs mv

soft link

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1)if primary removed ---> data cant be accessed

2) new inode

3) ln -s primaryfile softlinkfile

hard link

1)same inode

2) ln primary hardlink

Cp vs hard link

cp f1 f2

f1 --> b7 --> b9 (inode = 443)

f2 --> b50 --> b52 (inode == 999)

Each file is different f1 or f2

hard link

f1 --> b30 - b35(inode= 2)

f2 --> b30-b35 (inode = 2)

both files f1 or f2 contains same data

Ch:13
compress, archive
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compress (size)--> gzip, bzip, xz

gunzip, bunzip, unxz

archive --> tar

tar -cf archive.tar files+direct

tar -xf archive.tar -C /path/to/extract

compress and archive

gzip

tar -czvf backup.tar /etc/

bzib2

tar -cjvf backup.tar /etc/

xz

tar -cJvf backup.tar /etc/

tar -xzvf backup.tar

tar -xzf backup-new.tar -C specifice-path

Ch:14
packages

COMMAND	WHAT IT DOES	EXAMPLE
dnf install	Install package + dependencies	dnf install httpd
dnf remove	Remove/uninstall package	dnf remove httpd
dnf update	Update all or specific package	dnf update httpd
dnf check-update	List available updates (no install)	dnf check-update
dnf search	Search package by name/description	dnf search web server
dnf info	Show package details	dnf info httpd
dnf list installed	List all installed packages	dnf list installed

dnf provides	Find which package owns a file/cmd	dnf provides ifconfig
dnf group list	List available package groups	dnf group list
dnf group install	Install all packages in a group	dnf group install "Dev Tools"
dnf repolist	List enabled repositories	dnf repolist all
dnf history	Show past transactions	dnf history
dnf history undo	Undo a past transaction	dnf history undo 5
dnf module list	List modules and streams	dnf module list php
dnf module install	Install specific module stream	dnf module install php:8.2
dnf autoremove	Remove orphan dependencies	dnf autoremove
dnf downgrade	Roll back to older version	dnf downgrade httpd
rpm -qa	List all installed RPMs	rpm -qa | grep http
rpm -q	Check if package is installed	rpm -q httpd
rpm -ql	List files installed by package	rpm -ql httpd
rpm -qf	Which package owns this file?	rpm -qf /etc/hosts
rpm -qi	Show package info	rpm -qi httpd

ch15-access-file-system

TERM	MEANING
filesystem	Method of organizing data on a disk (ext4, xfs, vfat)

block device	Storage device — disk, partition, LVM volume
mount	Attach a filesystem to a directory (mount point)
mount point	Directory where the filesystem is attached
umount	Detach a mounted filesystem
UUID	Unique ID for a partition — safer than /dev/sda1
label	Human-readable name assigned to a partition
/etc/fstab	Permanent mount config — mounts at every boot
inode	Data structure storing file metadata (not name/data)
hard link	Another name pointing to same inode
soft link	Symbolic link — pointer to another file path

block devices:

hda --> hard disk

sda --> sata, sas, usb

vda --> virtual disk

nvme0

nvme1 ---> ssd

Lisk blocks:

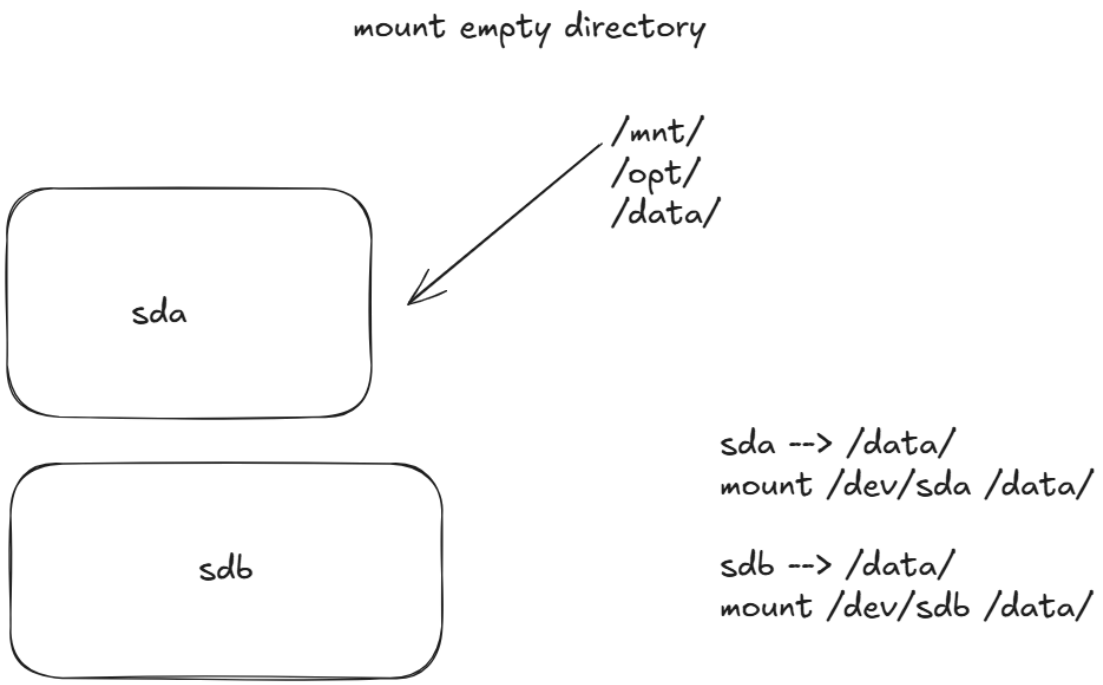
lsblk

List UUID:
blkid
temperoray:
mount /dev/sdb data/
unmount data/ or unmount /dev/sdb or unmount uuid

permemnant:
vim /etc/fstab
UUID=e5f6-g7h8 /data xfs defaults 0 0
or
/dev/sdb1 /data xfs defaults 0 0

COMMAND	KEY OPTIONS	WHAT IT DOES	EXAMPLE
lsblk	-f	List block devices as tree	lsblk -f
blkid	/dev/sda1	Show UUID and filesystem type	blkid /dev/sda1
df -h	-hT	Disk space usage (human readable)	df -hT
du -sh	-sh *	Directory size	du -sh /var/log/*
mount	-t -o ro/rw	Mount a filesystem	mount /dev/sdb1 /mnt/data
mount -a	—	Mount all entries in /etc/fstab	mount -a
umount	-l -f	Unmount a filesystem	umount /mnt/data
lsof	/mountpoint	Show open files on mount	lsof /mnt/data

fuser -m	-m -k	Show/kill PIDs using mount	fuser -m /mnt/data
ln	—	Create hard link	ln file.txt hardlink.txt
ln -s	-s	Create symbolic (soft) link	ln -s /etc/httpd conf
find	-name -user -size -type -perm	Search for files	find / -name "*.conf"



filesystem:

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xfs, ext1--4, vfat

retrieve, store disk

disk --> raw

disk + filesystem

mount --> temp(mount,nmount), persistent(/etc/fstab)

LAB 04

1. List the user commands and redirect the output to /tmp/commands.list
2. Count the number of user commands
3. Get all the users names whose first character in their login is 'g'.
4. Get the logins name and full names (comment) of logins starts with "g".
5. Save the output of the last command sorted by their full names in a file.
6. Display the number of users who is logged now to the system.
7. Display lines 7 to line 10 of /etc/passwd file
8. What happens if you execute:
    a. cat filename1 | cat filename2
    b. ls | rm
    c. ls /etc/passwd | wc –l

LAB 04 – Processes

1. Issue the command sleep 100.
2. Stop the last command.
3. Resume the last command in the background.
4. Issue the jobs command and see its output.
5. Send the sleep command to the foreground and send it again to the background.
6. Kill the sleep command.
7. Display your processes only.
8. Display all processes except yours.
9. Use the pgrep command to list your processes only.

10. Kill your processes only.

LAB 04: Exploring the Logging System

1. Display the last 20 lines of the system log file /var/log/messages.
2. Display the last 10 lines of /var/log/secure and explain what this file contains.
3. Use the tail -f command to monitor /var/log/messages in real time. Open another terminal and log in/out, then observe the output.
4. What is the difference between /var/log/messages and /var/log/dmesg?

Using journalctl

5. Display all journal log entries using journalctl.
6. Display only kernel messages using journalctl.
7. Display logs from the current boot only.
8. Display logs from the previous boot.
9. Filter journal logs to show only error and critical priority messages.
10. Display logs generated in the last 30 minutes.
11. Display logs for a specific service (e.g., sshd):

LAB 05

1. Compress a file by gzip and bzip2 commands and decompress it again. State the differences between two commands.
2. What is the command used to view the content of a compressed file.
3. Backup /etc directory using tar utility.
4. Starting from your home directory, find all files that were modified in the last two day.
5. Starting from /etc, find files owned by root user.
6. Find all directories in your home directory.
7. Write a command to search for all files on the system that, its name is ".profile".
8. Identify the file types of the following: /etc/passwd, /dev/pts/0, /etc, /dev/sda
9. List the inode numbers of /, /etc, /etc/hosts.

10. Copy /etc/passwd to your home directory, use the command diff and Edit in the file you copied, and then use the command again, and check the output.
11. Create a symbolic link of /etc/passwd in /boot.
12. Create a hard link of /etc/passwd in /boot. Could you? Why?
13. Add disk and create a mount point to empty directory with 2 ways temporary and permenant